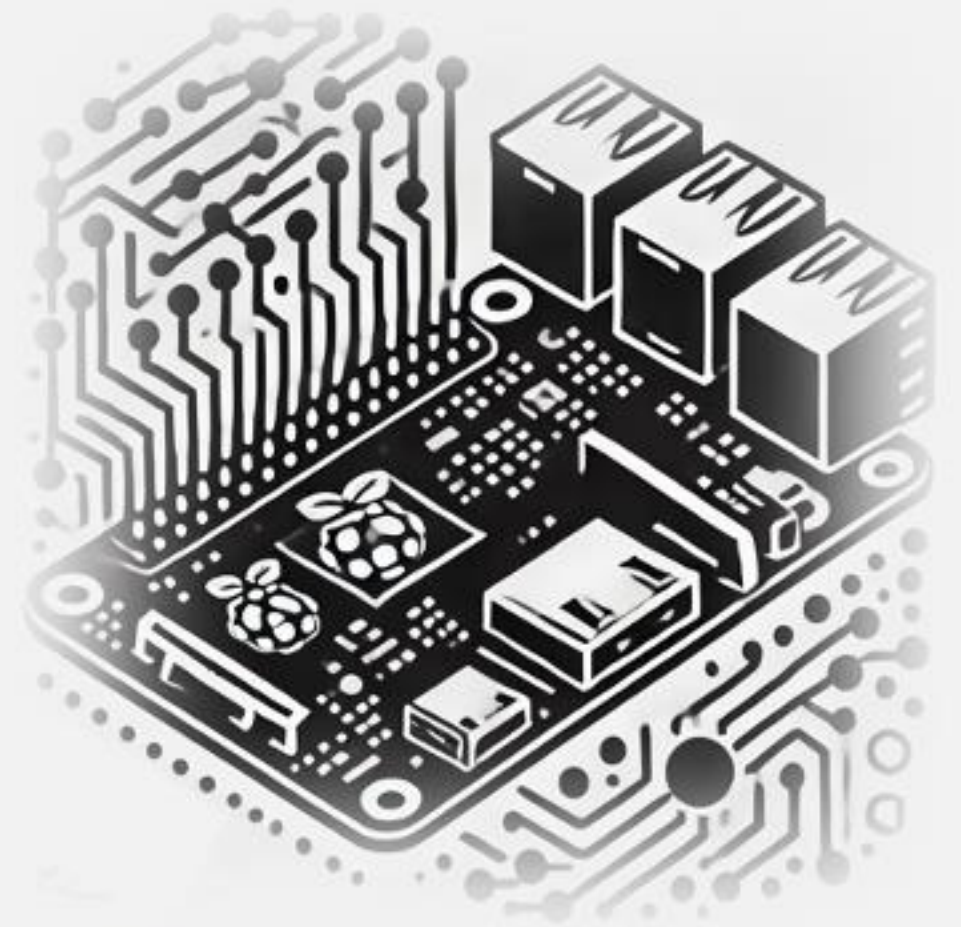


IESTIO5 – Edge AI

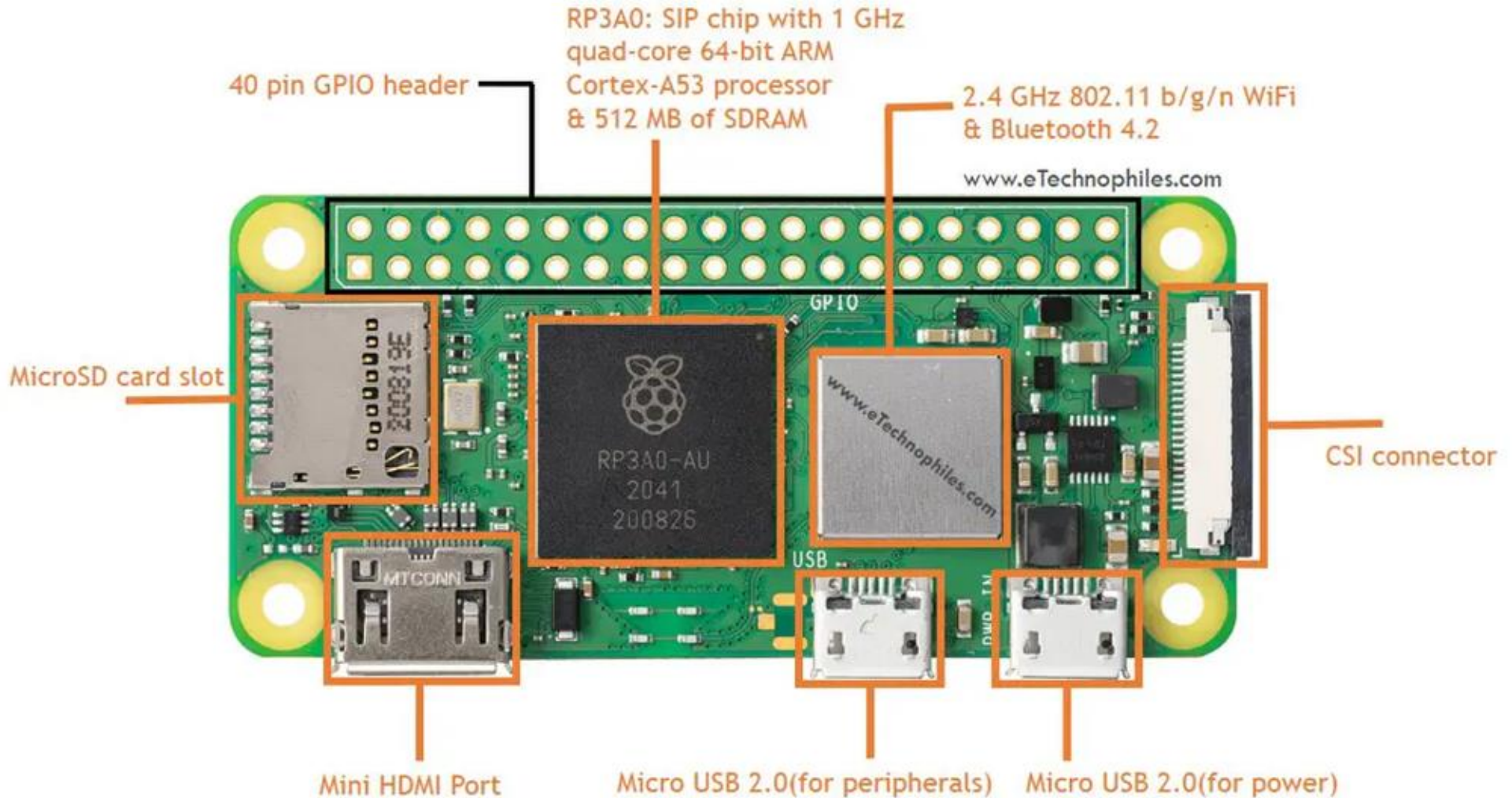
Machine Learning System Engineering

3. Raspberry Pi Setup Walkthrough

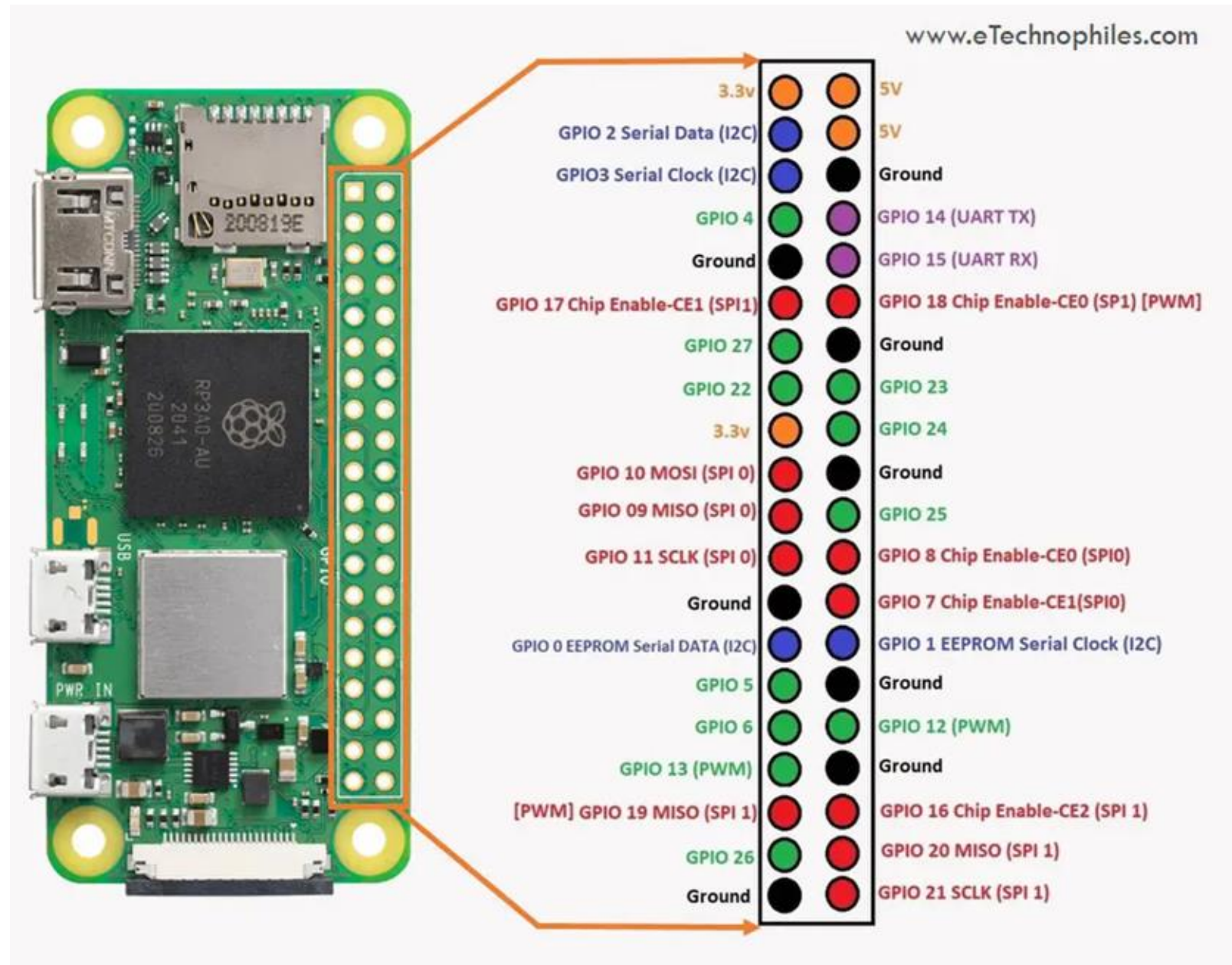


Raspberry Pi W2

Board Layout



GPIO pinout



Installing the OS

Use Raspberry Pi Imager

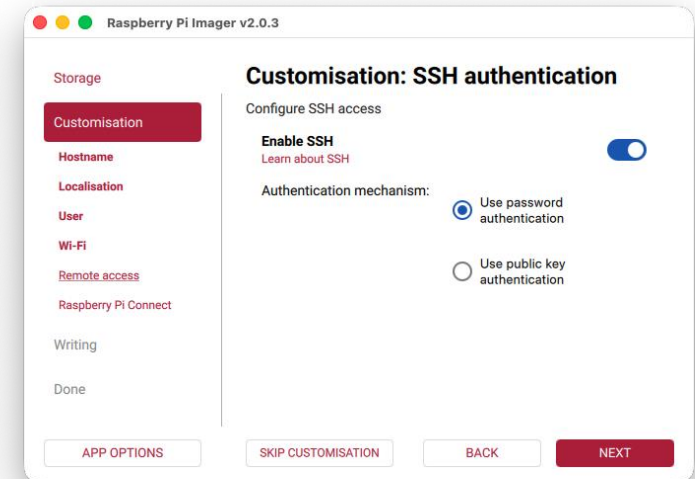
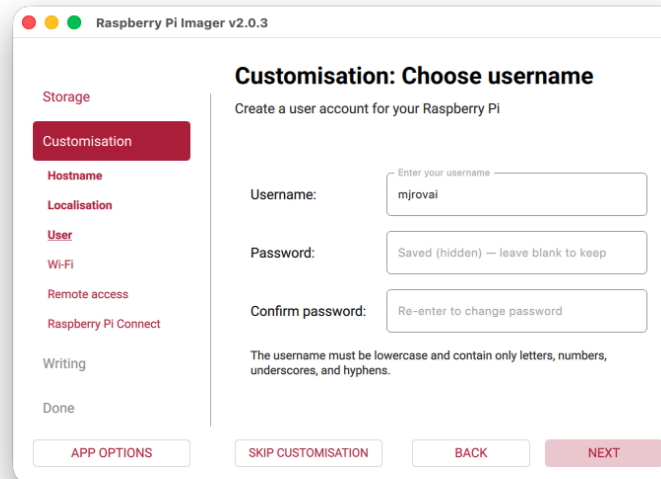
<https://www.raspberrypi.com/software/>

and select:

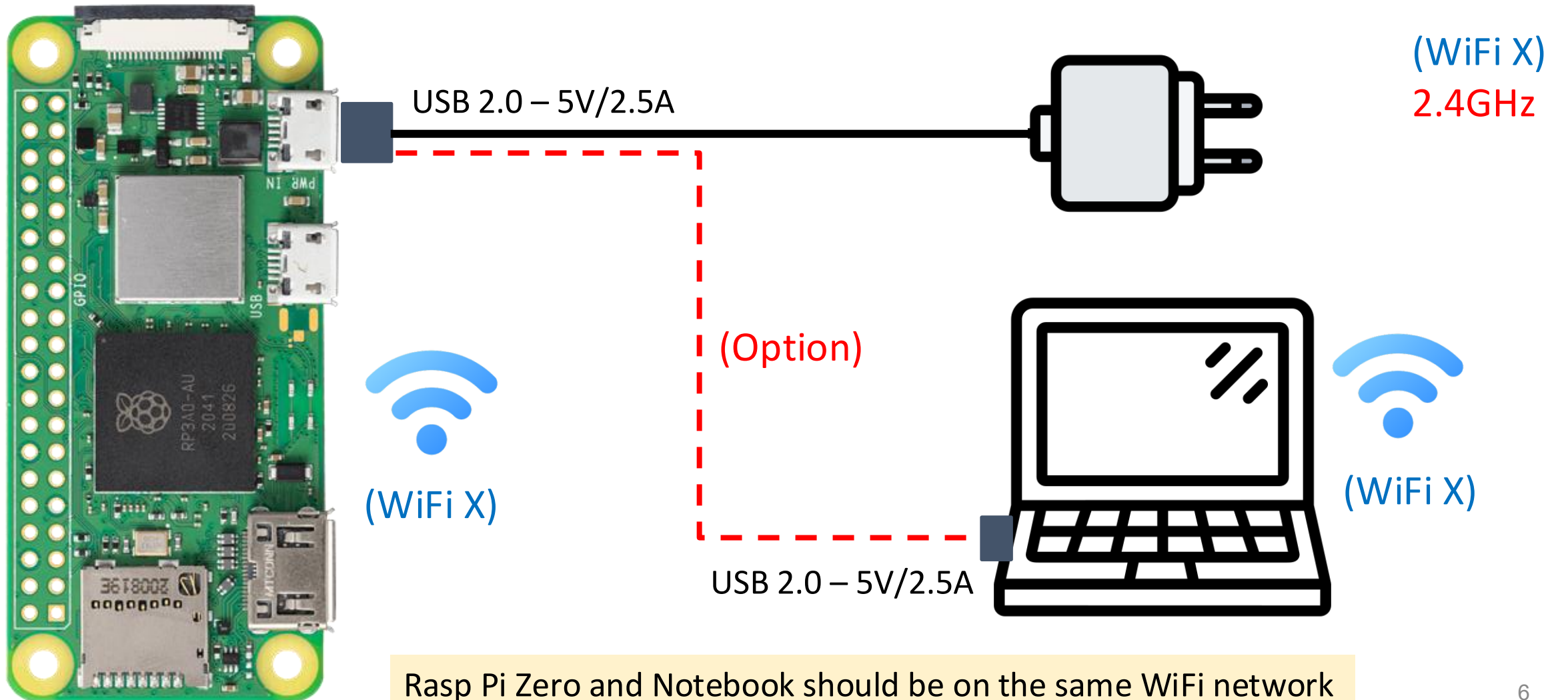
- RASPBERRY PI ZERO 2W
- RASPBERRY PI OS LITE (64-BIT)

Headless setup: enable SSH, Wi-Fi config

- Define **hostname**, **username**, and **password**

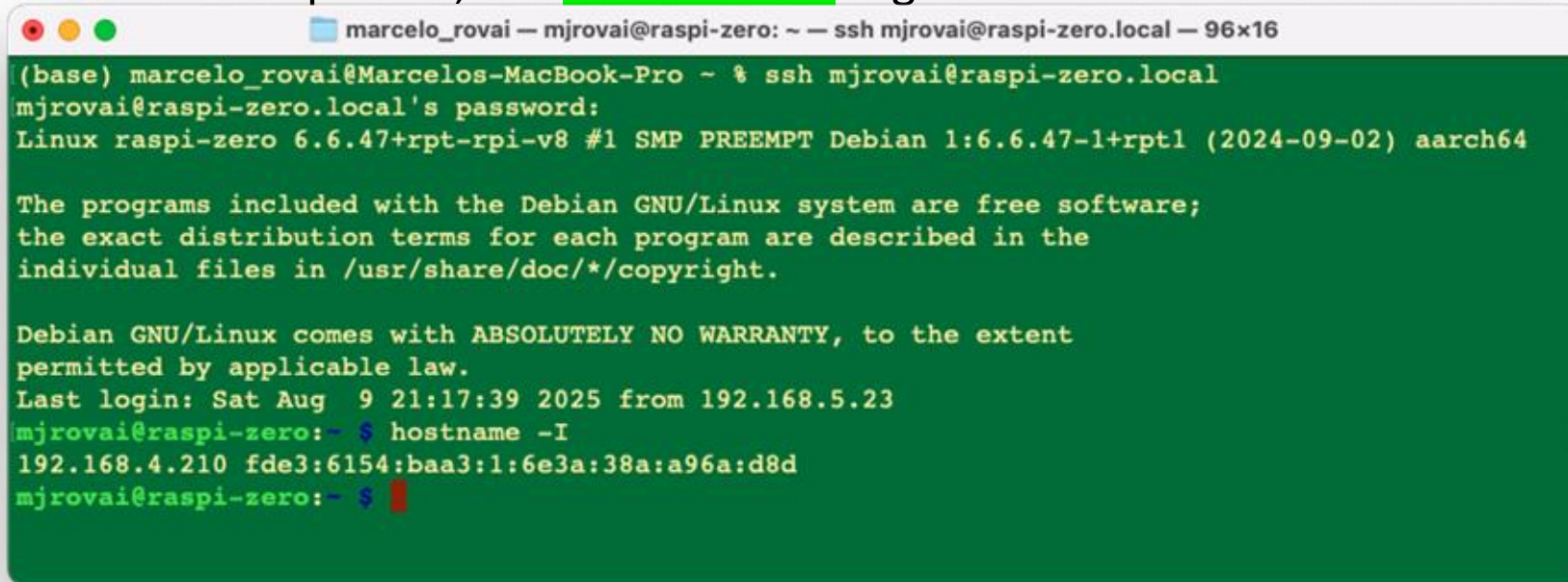


Power Supply and WiFi Network



Connecting to the Pi – SSH Via Terminal

- Enter with: `ssh username@hostname.local` (i.e., `mjrovai@raspi-zero.local`)
 - Once in the Raspi-Zero, use `hostname -I` to get the IP Address



```
marcelo_rovai — mjrovai@raspi-zero: ~ — ssh mjrovai@raspi-zero.local — 96x16
(base) marcelo_rovai@Marcelos-MacBook-Pro ~ % ssh mjrovai@raspi-zero.local
mjrovai@raspi-zero.local's password:
Linux raspi-zero 6.6.47+rpt-rpi-v8 #1 SMP PREEMPT Debian 1:6.6.47-1+rpt1 (2024-09-02) aarch64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat Aug  9 21:17:39 2025 from 192.168.5.23
mjrovai@raspi-zero:~$ hostname -I
192.168.4.210 fde3:6154:baa3:1:6e3a:38a:a96a:d8d
mjrovai@raspi-zero:~$
```

Note: On Windows, use Command Prompt (cmd) or PowerShell.

- Knowing the IP address: Enter with:
`ssh username@ip_address` (i.e., `mjrovai@ 192.168.4.210`)

Initial Linux Commands

1. Package mgmt:

- `sudo apt update`
- `sudo apt upgrade`
- `sudo reboot`

2. System: (Open in a new terminal window)

- `htop`

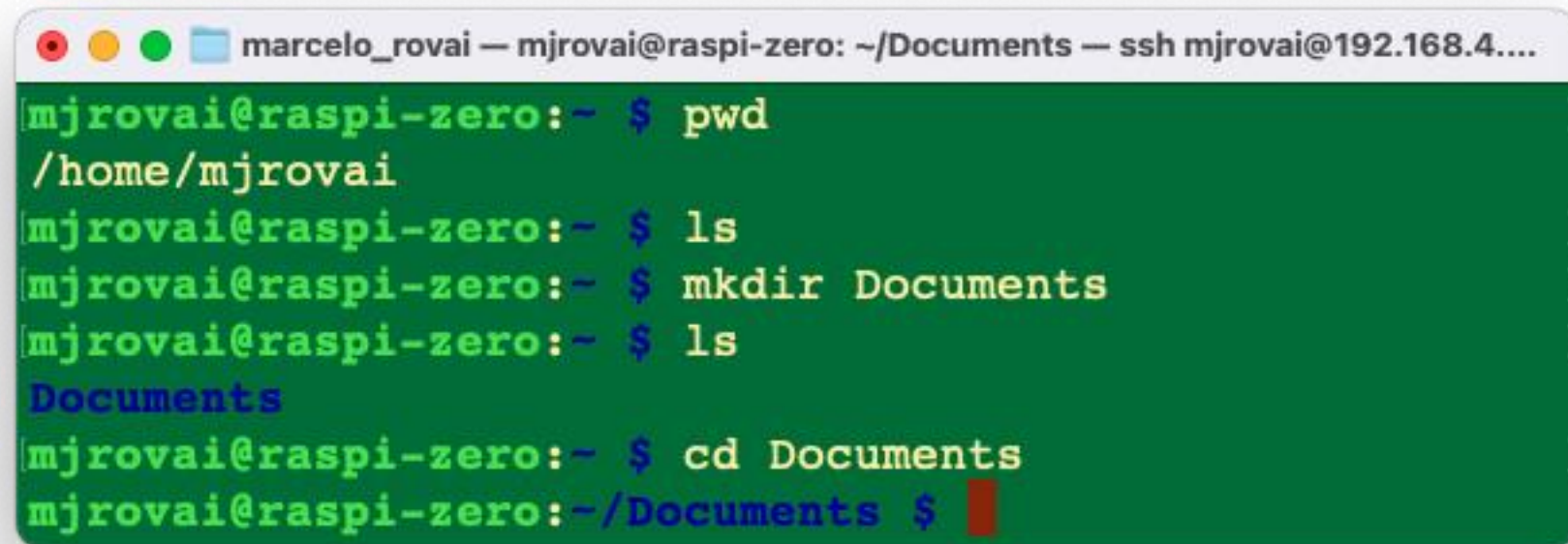
```
marcelo_rovai — mjrovai@raspi-zero2w-26: ~ — ssh mjrovai@raspi-zero2w-26.local — 80x24

0[ 0.0%] Tasks: 25, 4 thr, 104 kthr; 1 running
1[ 0.7%] Load average: 0.05 0.01 0.00
2[ 0.0%] Uptime: 01:27:26
3[ 0.0%]
Mem[||||||| |||] 94.5M/415M
Swp[ 0K/415M]

Main I/O
PID USER PRI NI VIRT RES SHR S CPU% MEM% TIME+ Command
1150 mjrovai 20 0 7248 3740 3092 R 0.7 0.9 0:00.30 htop
1 root 20 0 25092 14352 10432 S 0.0 3.4 0:08.30 /sbin/init
273 root 20 0 28748 8404 7140 S 0.0 2.0 0:00.57 /usr/lib/syst
314 systemd-ti 20 0 92244 7820 6700 S 0.0 1.8 0:00.32 /usr/lib/syst
324 root 20 0 35064 10056 7592 S 0.0 2.4 0:00.72 /usr/lib/syst
346 systemd-ti 20 0 92244 7820 0 S 0.0 1.8 0:00.01 /usr/lib/syst
515 avahi 20 0 5840 3424 2968 S 0.0 0.8 0:04.45 avahi-daemon:
516 root 20 0 12884 6224 5636 S 0.0 1.5 0:00.20 /usr/libexec/
517 root 20 0 6988 2764 2524 S 0.0 0.7 0:00.06 /usr/sbin/cro
518 messagebus 20 0 8480 4456 3532 S 0.0 1.0 0:01.56 /usr/bin/dbus
526 root 20 0 18856 8460 7272 S 0.0 2.0 0:00.49 /usr/lib/syst
535 avahi 20 0 5664 1652 1404 S 0.0 0.4 0:00.00 avahi-daemon:
546 root 20 0 333M 19880 16780 S 0.0 4.7 0:01.65 /usr/sbin/Net
F1Help F2Setup F3Search F4Filter F5Tree F6SortBy F7Nice -F8Nice +F9Kill F10Quit
```

Linux: Basic Commands

- `clear` -> Clear the terminal
- `pwd` -> Show the current directory: `/home/mjrovai`
- `ls` -> Lists the current directory content: (empty)
- `Mkdir <name>` -> Creates a directory: `mkdir Documents`
- `cd <dir>` -> Change to a directory `cd Documents`



```
marcelo_rovai — mjrovai@raspi-zero: ~/Documents — ssh mjrovai@192.168.4....  
[mjrovai@raspi-zero:~ $ pwd  
/home/mjrovai  
[mjrovai@raspi-zero:~ $ ls  
[mjrovai@raspi-zero:~ $ mkdir Documents  
[mjrovai@raspi-zero:~ $ ls  
Documents  
[mjrovai@raspi-zero:~ $ cd Documents  
mjrovai@raspi-zero:~/Documents $
```

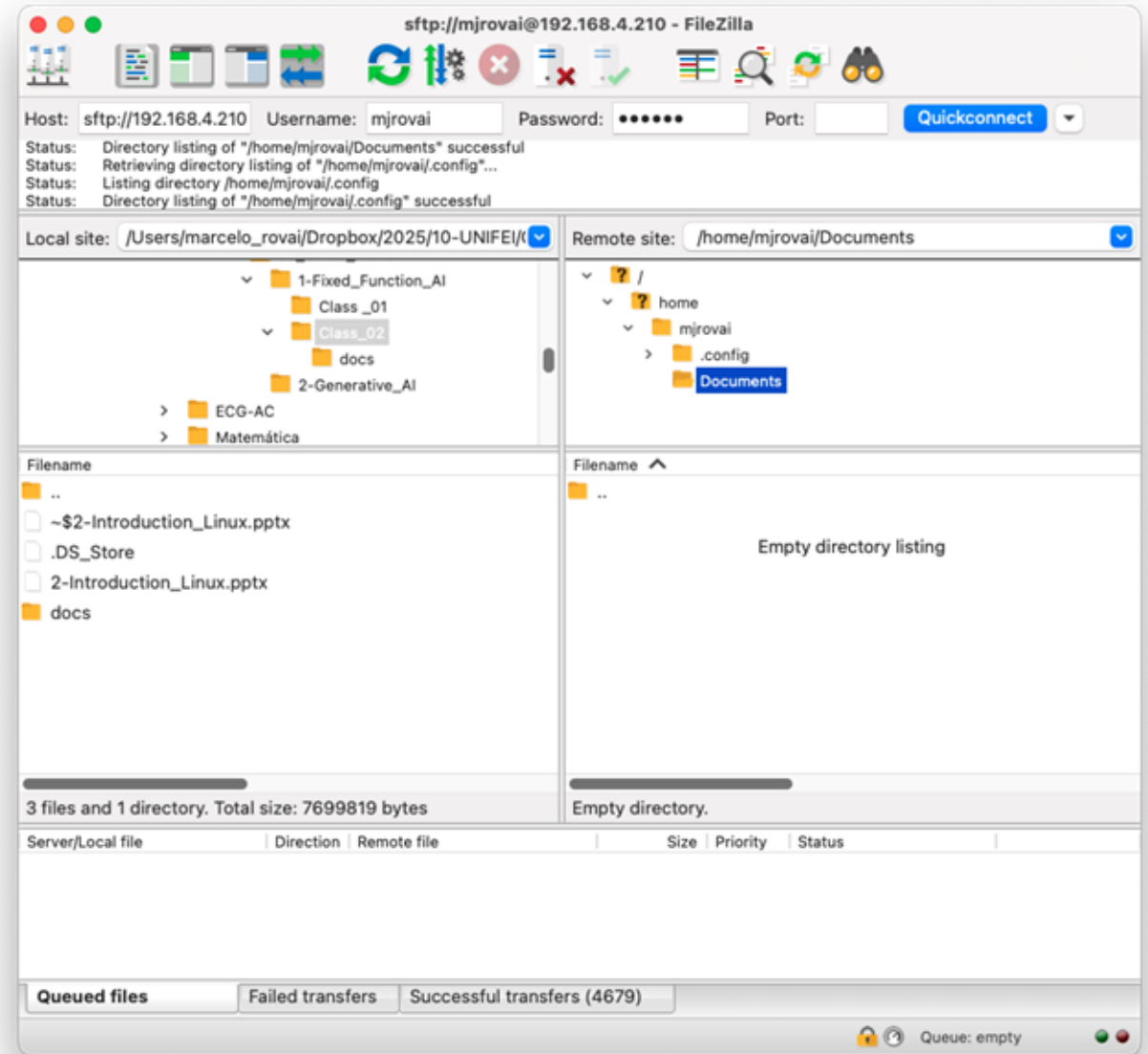
Using the Camara Module

1. Install camera software (if not pre-installed):
`sudo apt-get install libcamera-apps`
2. List the installed cameras:
`rpikam-hello --list-cameras`
3. Capture a 640x480 JPEG image:
`rpikam-jpeg --output test_cli_camera.jpg --width 640 --height 480`
4. Use the command `ls` to check if the image was saved in the current directory and transfer it to your desktop with FileZilla.



Transferring files using FTP

1. Install FileZilla Client in the Desktop
<https://filezilla-project.org/download.php?type=client>
2. Enter with Host Credentials
(i.e., sftp://192.168.4.210)



Tips

- Connecting the Raspberry Pi to the Computer via USB can be unstable for heavy use. Prefer a 5V/2.5 Power Supply (same as used for mobile phones)
- The WiFi Network should be 2.4GHz
- Always turn off the Raspberry Pi, using the command:
`sudo shutdown -h now`
- Install packages using
`sudo apt install <package name>`

Questions?

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rovai@unifei.edu.br



UNIFEI